NORTH SOUTH UNIVERSITY

# Arterial–Peripheral Pulse Propagation Model for Fingertip PPG Signal Generation

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## OBJECTIVE

To simulate fingertip photoplethysmographic (PPG) waveforms using a multi-stage arterial–peripheral model linking blood flow dynamics, vascular compliance, pulse-wave propagation, and optical absorption mechanisms.

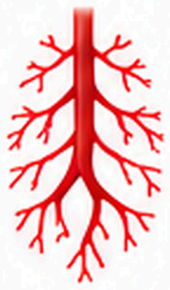
## 1. BACKGROUND & MOTIVATION

- Photoplethysmography (PPG) is the basis of pulse oximetry and a key signal in wearable healthcare.
- Realistic PPG simulation requires a model that reflects the physiological generation and propagation of arterial pulse.
- We present an arterial–peripheral pulse propagation model that reproduces key features of the pulse oximeter waveform.



Pulse Oximeter


Fingertip PPG Measurement


Peripheral Arterial Network

## 2. THEORETICAL FOUNDATION: FROM FLUID DYNAMICS TO PPG

### 2.1 Navier–Stokes Equations

Incompressible fluid

Continuity equation:

$$\nabla \cdot u = 0$$

Momentum equation:

$$\rho\left(\frac{\partial u}{\partial t} + (u \cdot \nabla)u\right) = -\nabla P + \mu \nabla^2 u$$

Where:
$u$ : velocity vector (m/s)
$P$ : pressure (Pa)
$\rho$ : density (kg/m³)
$\mu$ : dynamic viscosity (Pa·s)

### 2.2 Womersley Arterial Flow

Pulsatile flow in a cylindrical artery



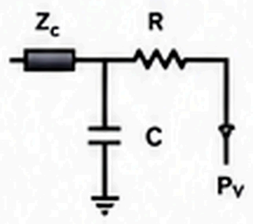
Womersley number:

$$\alpha = R\sqrt{\frac{\omega\rho}{\mu}}$$

where $\omega = 2\pi f$
($f$ : cardiac frequency)

Accounts for inertial and viscous effects in pulsatile flow.

### 2.3 Windkessel Peripheral Model (3-Element)




Represents characteristic impedance ($Z_c$), peripheral resistance (R) and compliance (C).

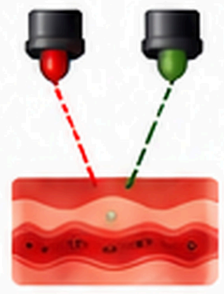
Models pressure damping and diastolic decay in the peripheral vasculature.

### 2.4 Microvascular Volume Change



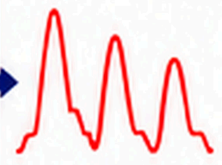
Peripheral pressure drives pulsatile blood volume variation in the microvasculature.

### 2.5 Optical Transduction



Volume change modulates light absorption according to the Beer–Lambert law to produce the PPG signal.

### 2.6 PPG Signal



Simulated pulse oximeter–like PPG waveform.

### Key Equations in Theoretical Foundation

**Womersley arterial transmission (Frequency domain):**

$$H_W(\omega) = \frac{e^{-j\omega\tau_a}}{1 + j\omega\tau_W}$$

$\tau_a$ : arterial delay.
$\tau_W$ : Womersley time constant.

**Windkessel (time domain):**

$$C\frac{dP_c(t)}{dt} = \frac{P_a(t) - P_c(t)}{Z_c} - \frac{P_c(t) - P_v}{R}$$

$$Q(t) = \frac{P_a(t) - P_c(t)}{Z_c}$$

$P_c(t)$ : peripheral pressure, $P_v$ : venous pressure

**Microvascular volume:**

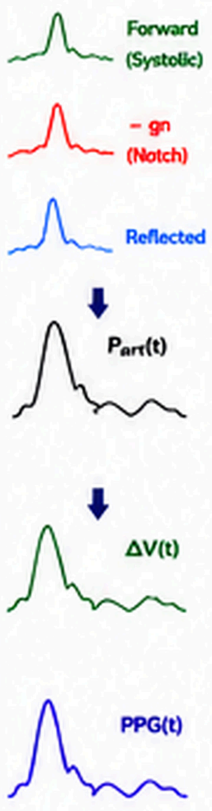
$$\Delta V(t) = C_{micro}\left[P_c(t) - P_{vef}\right]$$

**Optical model (Beer–Lambert inspired):**

$$PPG(t) = I_0\, e^{-k\tanh(\beta\, \Delta V(t))}$$

$I_0$ : incident light intensity, $k$ : optical gain, $\beta$ : nonlinearity

## 3. IMPLEMENTED ARTERIAL–PERIPHERAL–PPG MODEL (MATLAB)




### 1. Arterial Pulse Generation (Morphology Construction)

$$P_{art}(t) = g_2(t) - g_n(t) + g_2(t)$$

$$g_i(t) = A_i \exp\left(-\frac{(t - t_i)^2}{2\sigma_i^2}\right)$$

$i = 1$ (systolic), n (notch), 2 (reflected)



### 2. Peripheral Pulse Transformation

$$P_{per}(t) = 0.9\ S\{P_{art}(t - \tau_d)\}$$

- $\tau_d = 0.12$ s (propagation delay)
- $S(\cdot)$: Gaussian smoothing (mild)
- Amplitude attenuation: 10%



### 3. Microvascular Compliance

$$\Delta V(t) = C_{micro}\ P_{per}(t)$$

$$C_{micro} = 1.2$$



### 4. Optical Transformation (Beer–Lambert Inspired)

$$PPG(t) = e^{-0.7\tanh(1.5\Delta V(t))}$$

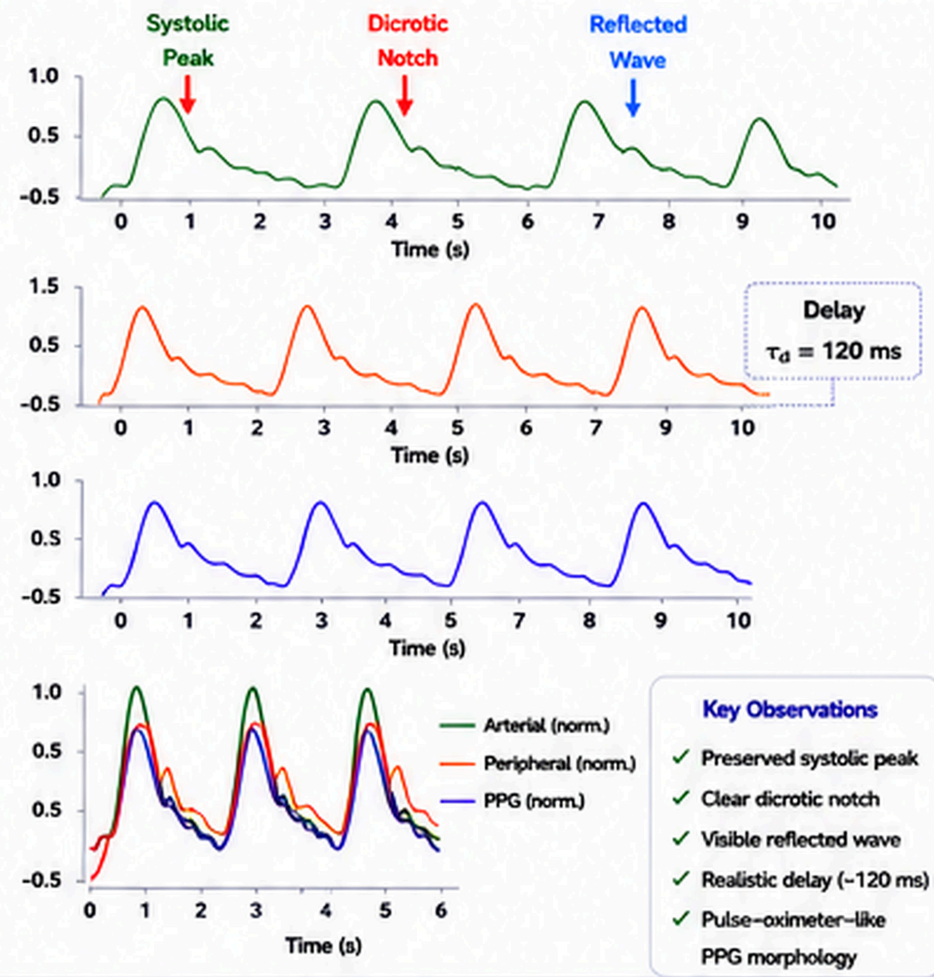
$(k = 0.7,\quad \beta = 1.5)$



### 5. Simulated PPG Signal (oximeter–like waveform)



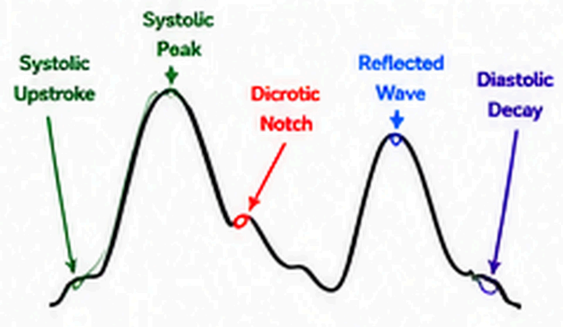
## 4. SIMULATION RESULTS (MATLAB)




**Key Observations**

- ✓ Preserved systolic peak
- ✓ Clear dicrotic notch
- ✓ Visible reflected wave
- ✓ Realistic delay (~120 ms)
- ✓ Pulse-oximeter–like PPG morphology

## 5. MORPHOLOGICAL FEATURES PRESERVED




The model reproduces key PPG-like morphology.

## 6. INTERPRETATION



### Arterial Pulse

- Generated by ventricular ejection.
- Notch occurs due to aortic valve closure.
- Reflected wave arises from impedance mismatch in the arterial network.



### Peripheral Pulse

- Propagates through arterial tree with delay.
- Damping and smoothing occur due to vascular resistance and compliance.



### PPG Signal

- Proportional to pulsatile blood volume in the microvasculature.
- Optical absorption changes produce the pulse oximeter waveform.

## 7. RESULTS / KEY FINDINGS

- The model reproduces major fingertip PPG morphological landmarks.
- Peripheral propagation introduces delay, attenuation, and waveform smoothing.
- Dicrotic notch and reflected-wave components can be preserved in the simulated PPG.
- The framework provides a physiologically interpretable basis for future PEMF–PPG modelling.

## 8. FUTURE WORK



Extend the model to compare baseline vs PEMF-induced waveform changes.



Estimate model parameters from experimental Doppler–PPG data under different PEMF configurations.

## 9. REFERENCES

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